Lecture 15

Integral Sliding Mode Control

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Sliding mode control and its application

Outline

Integral Sliding Mode

Consider the LTI system

$$\dot{x}(t) = Ax(t) + Bu(t) + Bd, \quad x(0) = x_0$$

where $|d| \leq d_{\max}$ and $u \in \mathbb{R}$.

Design an integral sliding manifold as

$$s(t) = G\left[x(t) - x_0 - \int_0^t (Ax(\tau) + Bu_0)\mathrm{d}\tau\right]$$

where $G \in \mathbb{R}^{1\times n}$ is projection matrix. Note that $s(t)|_{t=0} = 0$.

Design

$$u(t) = u_0 + u_1$$

where $u_1 = -K\operatorname{sign}(GBs(t))$ and $K > d_{\max}$.

Remark

- Here $x_0 + \int_0^t (Ax(\tau) + Bu_0)\mathrm{d}\tau$ is the nominal trajectory of the system.
- The sliding variable $s(t)$ is the measure of the deviation of the plant evolution with respect to the nominal trajectory.

Integral Sliding Mode

In control, u, nominal control u_0 can be designed as state feedback, LQR, PID, etc., irrespective of control component u_1.

Now differentiating $s(t)$ gives,

$$\begin{aligned}\dot{s}(t) &= G\left[Ax(t) + Bu(t) + Bd - Ax(t) - Bu_0\right] \\ &= G\left[Ax(t) + Bu_0 + Bu_1 + Bd - Ax(t) - Bu_0\right] \\ &= GBu_1 + GBd.\end{aligned}$$

Since $u_1 = -K\mathrm{sign}(GBs(t))$, using Lyapunov function $V = \frac{1}{2}s^2$ we obtain

$$\begin{aligned}\dot{V} &= s(t)\dot{s}(t) \\ &= -KGBs(t)\mathrm{sign}(GBs(t)) + s(t)GBd \\ &\leq -K|GBs(t)| + |GBs(t)|d_{\max} \\ &= -|GBs(t)|(K - d_{\max}).\end{aligned}$$

Note that $V|_{t=0} = 0$ and $\dot{V} < 0$, hence $s(t) = 0$ for all $t \geq 0$.

ISM as Disturbance Observer

From the above, we see that control achieves sliding mode from the very beginning. This implies that the disturbances are also rejected from the initial time.

$$\begin{aligned}\dot{s} &= GBu_1 + GBd \\ &= -GBK\,\mathrm{sign}(GBs(t)) + GBd.\end{aligned}$$

To calculate the equivalent control, make $\dot{s} = 0$, then

$$u_{1_{\mathrm{eq}}} = -d.$$

In other words, $K\,\mathrm{sign}(GBs(t))\big|_{\mathrm{eq}} = d$. This concludes that, on the surface, the disturbance is completely rejected.

This is to be noted that when $s(t) = 0$, the control u_1 acts as the disturbance observer.

ISM for Nonlinear System

Consider a nonlinear system defined as

$$\dot{x} = f(x,t) + Bu(x,t) + \phi(x,t) \tag{1}$$

where $x \in R^n$, $u(x,t) \in R^m$ and $\phi(x,t)$ is perturbation due to model uncertainties or external disturbances.

Assumptions

1. rank $(B) = m$
2. Assume $\phi(x,t) = B\phi_m(x,t)$ and $\|\phi(x,t)\| \leq \|B\|\phi_{m_{\max}}$ for all $x,\ t$

Structure of the control law: In integral sliding mode, the control law is defined as

$$u(x,t) := u_0(x,t) + u_1(x,t) \tag{2}$$

- $u_0(x,t)$ is a nominal controller which can be designed for the nominal system (i.e., system without perturbation) by any method, i.e., PID, pole-placement, LQR, MPC.
- $u_1(x,t)$ is a discontinuous control law which rejects the matched perturbation and ensures sliding mode.

Sliding Manifold

The sliding manifold is defined as

$$s(x,t) := G\left[x(t) - x(t_0) - \int_{t_0}^{t} (f(x,\tau) + Bu_0(x,\tau))d\tau\right] \quad (3)$$

where $G \in R^{m \times n}$ is a projection matrix and chosen such that GB is invertible.

The term

$$x(t_0) + \int_{t_0}^{t} (f(x,\tau) + Bu_0(x,\tau))d\tau$$

can be thought of as a trajectory of the system, obtained by the nominal control $u_0(x,t)$, in the absence of perturbation term $\phi(x,t)$.

Function s(x,t) can be seen as[1] [2],

$$s(x,t) := Gx(t) - G\underbrace{\left[x(t_0) + \int_{t_0}^{t} (f(x,\tau) + Bu_0(x,\tau))d\tau\right]}_{\text{Nominal Trajectory}}$$

[1] V. I. Utkin, J. Shi, "Integral sliding mode in systems operating under uncertainty conditions", In the Proc. of Int. Conf. Decision and Control, Japan, 1996.

[2] F. Castaños, L. Fridman, "Analysis and design of integral sliding manifolds for systems with unmatched perturbations", IEEE Trans. Autom. Control, vol. 51, no. 5, pp. 853–858, 2006.

Existence Condition of Sliding Mode

We can consider $s(x,t)$ as a penalizing factor of the difference between actual and nominal trajectories projected along G.

Note: At $t = t_0$, $s(x, t_0) = 0$, so the system always starts at the sliding manifold. **NO REACHING PHASE**

State trajectory should be on the surface from the beginning and under the influence of disturbances. Let this control law be

$$u_1(x,t) = -\rho(x,t)\frac{(GB)^T s(x,t)}{\|(GB)^T s(x,t)\|} \tag{4}$$

Since the uncertainty is matched i.e., $\phi(x,t) = B\phi_m(x,t)$

$$\begin{aligned}\dot{s} =& G(\dot{x} - f(x,t) - Bu_0) \\ =& G(f(x,t) + Bu(x,t) + B\phi_m(x,t) - f(x,t) - Bu_0) \\ =& GBu_1(x,t) + GB\phi_m(x,t)\end{aligned}$$

using (4), $\dot{s} = -\rho(x,t)GB\frac{(GB)^T s(x,t)}{\|(GB)^T s(x,t)\|} + GB\phi_m(x,t)$

Existence Condition of Sliding Mode

Using Lyapunov function $V = \dfrac{1}{2}s^\top s$, we can find $\dot{V}$ as follows

$$
\begin{aligned}
\dot{V} = s^T \dot{s} &= -s^T \rho(x,t) GB \frac{(GB)^T s(x,t)}{\|(GB)^T s(x,t)\|} + s^T GB \phi_m(x,t) \\
&\leq -\rho(x,t)((GB)^T s(x,t))^T \frac{(GB)^T s(x,t)}{\|(GB)^T s(x,t)\|} + \|s^T(x,t) GB \phi_m(x,t)\| \\
&\leq -\rho(x,t) \frac{\|(GB)^T s(x,t)\|^2}{\|(GB)^T s(x,t)\|} + \|s^T(x,t) GB\| \|\phi_m(x,t)\| \\
&\leq -\rho(x,t) \|(GB)^T s(x,t)\| + \|s^T(x,t) GB\| \phi_{m_{\max}} \\
&= -\rho(x,t) \|(GB)^T s(x,t)\| + \|(GB)^T s(x,t)\| \phi_{m_{\max}} \\
&= -\|(GB)^T s(x,t)\| (\rho(x,t) - \phi_{m_{\max}}) \\
&< 0.
\end{aligned}
$$

where $\rho(x,t) > \phi_{m_{\max}}$.

Unmatched perturbation

Before we go for the analysis of unmatched perturbations, we study the following proposition.

Proposition : For any matrix $B \in R^{n\times m}$ with $rank(B) = m$ the following identity holds

$$I_n = BB^{+} + B^{\perp}{B^{\perp}}^{+} \tag{5}$$

where the columns of $B^{\perp} \in R^{n\times(n-m)}$ spans the null space of B^T and $B^{+} = (B^TB)^{-1}B^T$ is understood as the left inverse of B.

Proof: Consider a matrix $P = \begin{bmatrix} B^{+} \\ {B^{\perp}}^{+} \end{bmatrix}$. This matrix is nonsingular, since $P^{-1} = \begin{bmatrix} B & B^{\perp} \end{bmatrix}$.

We see that

$$PP^{-1} = \begin{bmatrix} B^{+}B & 0 \\ 0 & B^{\perp +}B^{\perp} \end{bmatrix} = \begin{bmatrix} I_m & 0 \\ 0 & I_{n-m} \end{bmatrix}.$$

Now reversing the order, we get $P^{-1}P = BB^{+} + B^{\perp}{B^{\perp}}^{+} = I_n$.

Decomposition of uncertainty

- Actual uncertainty can be decomposed as

$$\phi = \phi_m + \phi_u, \quad \phi_m := BB^+\phi, \quad \phi_u := B^\perp B^{\perp^+}\phi$$

- By considering this uncertainty, equivalent control can be obtained by setting $\dot{s} = 0$

$$u_{1eq} = -B^+\phi - (GB)^{-1}G\phi_u$$

- By substituting this u_{1eq} in (1)

$$\begin{aligned} \dot{x} &= f(x,t) + B(u_0 - B^+\phi - (GB)^{-1}G\phi_u) + BB^+\phi + B^\perp B^{\perp^+}\phi \\ &= f(x,t) + Bu_0 + [I - B(GB)^{-1}G]\phi_u \end{aligned} \tag{6}$$

- It can be seen that the matched perturbation term does not exist in the closed loop. While the unmatched term does remain.
- In the closed loop system (6), the unmatched perturbation is multiplied by the matrix

$$\Gamma := [I - B(GB)^{-1}G] \tag{7}$$

- Is there any way to deal with the influence of unmatched perturbations?

Decomposition of uncertainty

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- Is there any way to deal with the influence of unmatched perturbations?
  There is no way to nullify the influence of the unmatched perturbations with this technique. However, we should try to minimize Γ. To do so, we can use G as a free parameter.

Minimization of the influence of unmatched perturbation

Questions:
Is there any matrix G such that the norm of the unmatched term $(\Gamma\phi_u)$is minimal?

Does the matrix Γ amplify the unmatched perturbation?

Let $\phi_{eq} = \Gamma\phi_u$

Proposition: B^T is a matrix which minimizes $\|\phi_{eq}\|$
Proof:

$$\|\Gamma\phi_u\| = \|[I - B(GB)^{-1}G]\phi_u\|_2 = \|\phi_u - B\Omega\|_2 \tag{8}$$

where $\Omega := (GB)^{-1}G\phi_u$. Thus, (8) can be written as

$$\Omega^* = \arg\min_{\Omega\in R^m} \|\phi_u - B\Omega\|_2$$

According to projection theorem $\Omega^* = B^+\phi_u$ is the solution. Now, with $G = B^T$

$$\Omega = (B^T B)^{-1}B^T\phi_u = B^+\phi_u = \Omega^*.$$

The other possibility for the optimal choice of matrix G is B^+.

ISM as a Disturbance Observer

Consider the system (1) with matched perturbation. The discontinuous control law "effect" can be given by Equivalent Control law.

Consider as obtained earlier

$$\dot{s} = GBu_1(x,t) + GB\bar{\phi}(x,t) \tag{9}$$

From this equation, the equivalent control

$$u_{1eq} = -\bar{\phi}(x,t) \tag{10}$$

The control is exactly equal to the negative of disturbance. The actual discontinuous control (4) can be passed through a low pass filter to obtain the actual value of uncertainty.

Simulation Results

Consider the double integrator plant described by the following equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} (u + \rho) \tag{11}$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{12}$$

where ρ is disturbance.

Above equation is in standard form $\dot{x} = Ax + Bu$ and output $y = Cx$

Step-1: Design nominal controller u_0.

- Suppose the controller requirement is output tracking.
- To track constant trajectory, the nominal controller can be calculated as follows

$$u_0 = Fx + G_1 r. \tag{13}$$

- Using control u_0 in nominal system we obtain

$$\dot{x} = (A + BF)x + BG_1 r$$

Simulation Results

- Using Laplace transform, we get

$$X(s) = (sI - (A + BF))^{-1} BG_1 R(s)$$
$$Y(s) = C(sI - (A + BF))^{-1} BG_1 R(s)$$

- For DC gain, $s = 0$,

$$y(t) = -C(A + BF)^{-1} BG_1 r(t)$$

- We want $y(t) = r(t)$, this can be achieved by choosing

$$G_1 = -[C(A + BF)^{-1} B]^{-1}. \tag{14}$$

- By pole placement approach F can be calculated as follows F=$[-2 \quad -3]$
- From (14), $G_1 = 2$

Plant response with nominal controller u_0

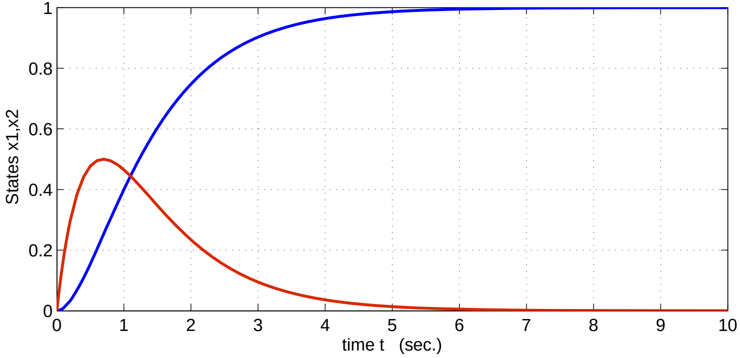
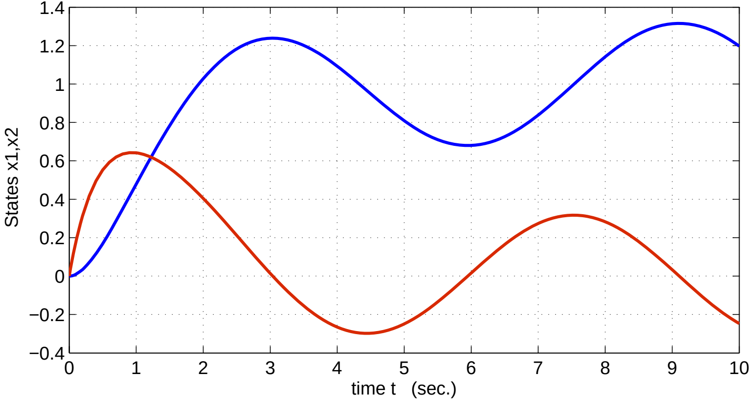


Figure:

Plant response with nominal controller u_0 and disturbance $\rho = sin(t)$



Design of Integral sliding mode controller

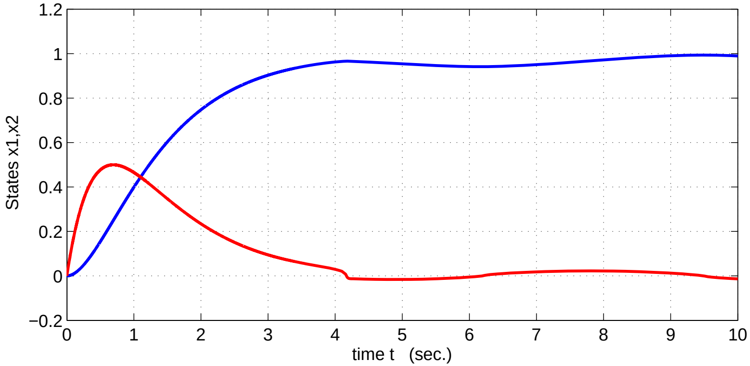
- **Step:2**
  Calculate sliding surface (3) with $G = B^{+} = [0 \quad 1]$

- **Step:3**
  Calculate control law.
  Control law can be calculated by choosing $K = 1.1$

- **Step:4**
  Form total control as

$$u = u_0 + u_1$$

Plant response with ISM controller ($u = u_0 + u_1$) and disturbance $\phi(x,t) = sin(t)$



Some Observations

- Nominal control u_0 decides plant dynamics.
- ISM controller u_1 works like a perturbation estimator.
- With ISM, we can combine two independent controllers, and the resultant controller ensures robust performance.

Thank You!